SOMAIYA
VIDYAVIHAR
K J Somaiya Institute of Technology
(Formerly known as K J Somaiya Institute of Engineering and Information Technology)
An Autonomous Institute Permanently Affiliated to University of Mumbai.

S.Y. Mini Project Presentation

AgriAdvisor: An El Niño-Aware RAG-based Agricultural Advisory System

Presented By (Group 11)

Panth Shah (RollNo: 31)

Vedant Shinde (Roll No: 35)

Hamood Siddiqui (Roll No: 36)

Project Guidance & Institution
Guided by

Prof. Bhagyashree Ingle

Department of AI & DS
K.J.SomaiyaInstituteofTechnology, Sion, Mumbai

Sr. No.	Content
1	Introduction
2	Literature Survey
3	Objective of Project
4	Implementation
5	Expected Results
6	Applications
7	Future Scope
8	Conclusion
9	References
10	Time Plan

INTRODUCTION

BACKGROUND

- **Domain:** Applied AI for climate-resilient agriculture.
- **Context:** El Niño disrupts the Indian monsoon, causing rainfall deficits and crop stress. Rainfed Maharashtra (Marathwada/Vidarbha) is highly exposed.
- **Objective:** Merge verified government guidance with AI to deliver localized, explainable advisories.

PROBLEM STATEMENT

- **The Gap:** ICAR & IMD publish reliable guidance, but it remains fragmented across multiple separate portals and documents.
- **Limitation:** No synthesized, reasoned recommendation exists for specific district, crop, and El Niño scenarios.

AgriAdvisor bridges this critical translation gap.

MOTIVATION

- **Urgency:** Extension workers advising farmers during tight sowing windows lack time to manually search and cross-reference government PDFs.
- **Impact:** A synthesized, traceable advisory tool accelerates trustworthy decision-making before critical sowing windows close.

OBJECTIVES

PRIMARY OBJECTIVE

To design and implement
a Retrieval-Augmented Generation (RAG) system thatsynthesizes verified agricultural guidance into localized, explainable advisories for KVK officers and extension workers during El Niño years.

SPECIFIC GOAL 01

Knowledge Base: To design/develop a knowledge base architecture integrating ICAR contingency plans, IMD advisories, and verified ENSO literature for four representative Maharashtra districts.

SPECIFIC GOAL 02

Retrieval Pipeline: To implement/analyze a retrieval pipeline that reprioritizes guidance based on the selected El Niño scenario intensity — Strong, Moderate, Weak, or Neutral.

SPECIFIC GOAL 03

Evaluation & Quality: To evaluate/optimize advisory quality by generating output in a traceable Recommendation → Reason → Source format, benchmarked against a plain LLM baseline with no retrieval.

LITERATURE REVIEW & RESEARCH GAP

Author(s) & Year	Paper Title	Method/Technique	Objective	Limitations	KeyFindings	Research Gap
Kumar, K.K. et al. (2006) — Science	Unraveling the Mystery of Indian Monsoon Failure During El Niño"	Statistical analysis of historical SST and monsoon rainfall correlation	Establish the link between El Niño and Indian monsoon drought	Doesn't address regional/temporal variability within India	Central-Pacific-centered El Niño events most strongly suppress Indian rainfall; severe droughts have historically coincided with El Niño	Pacific-centered El Niño events most strongly suppress Indian rainfall; severe droughts have historically coincided with El NiñoEstablishes the foundational link but not how it translates into localized, actionable guidance — the exact gap AgriAdvisor fills

PROPOSED METHODOLOGY

Proposed Approach
AgriAdvisorfollows a Retrieval-Augmented Generation (RAG) approach. Instead of relying on general training knowledge, the system retrieves verified content from a curated knowledge base, then generates a grounded, localized advisory.

1. INGESTION

Data Sources
Loads ICAR district contingency plans and IMD agromet advisories for four target districts.

2. PREPROCESSING

Semantic Chunking
Splits documents into semantic chunks and converts them into embeddings for meaning-based search.

3. RETRIEVAL

Contextual Match
Given a district, crop, month, and El Niño scenario, finds the specific chunks that actually apply — not the whole document.

4. GENERATION

LLM Synthesis
The LLM writes its answer using only the retrieved content, formatted in a strict structured flow:
Rec. → Reason → Source

SYSTEM ARCHITECTURE

1. INGESTION

- **Load ICAR:** District Contingency Plans
- **LoadIMD:** Agromet Advisory Bulletins

2. PROCESSING

- **Chunking:** Split documents into semantic segments
- **Embeddings:** Generate via Ollama nomic-embed-text

3. RETRIEVAL

- **Scenario Ranking:** Weighted by El Niño intensity (Strong/Moderate/Weak/Neutral)
- **Metadata:** District and crop filtering

4. GENERATION

- **LLM Synthesis:** Ollama, local inference
- **Structured Output:**

Recommendation→Reason → Source

TOOLS, TECHNOLOGIES & RESOURCES

HARDWARE

- **Processor:** Intel Core i5 / equivalent (no GPU required, local LLM via Ollama on CPU)
- **Memory:** 8GB+ RAM recommended for embedding & local inference
- **Storage:** Standard SSD for OS, Python env, and vector storage

SOFTWARE & STACK

- **Language:** Python
- **Frameworks:** LangChain, Ollama, ChromaDB, Streamlit (UI)
- **Database:** ChromaDB (vector database for semantic retrieval)

DATASETS

- **ICAR:** District Agricultural Contingency Plans (Latur, Beed, Yavatmal, Amravati)
- **IMD:** Agromet Advisory Bulletins for target districts
- **NOAA:** ONI classification reference for scenario selection
- **Literature:** Verified academic sources on ENSO–India relationships

WORK COMPLETED SO FAR

1. SCOPE & SITES

- **Project Definition:** Finalized v2 scope, primary users (KVK officers/extension workers), inputs, outputs, and explicit boundaries.
- **District Selection:** Selected and justified four representative districts (Latur, Beed, Yavatmal, Amravati) based on data-availability.

2. DATA COLLECTION

- **Contingency Plans:** Collected official ICAR plans for all four districts from the ICAR-CRIDA repository.
- **Advisory Bulletins:** Collected IMD Agromet Advisory Bulletins for all target districts.
- **Lit Verification:** Verified 3 of 11 literature sources at full citation level (DOI, journal, and authorship).

3. ANALYSIS & GAP DESIGN

- **Requirements Matrix:** Built matrix mapping user questions to sources; confirmed ICAR/IMD cover most operational needs.
- **Architecture Gap:** Identified and designed a scenario-based retrieval policy for El Niño intensity (implementation pending).

FINAL / EXPECTED OUTCOMES

SYSTEM & ARCHITECTURE

- **Core Functionality:** RAG-based decision support system for KVK officers and extension workers. [cite: 10]
- **Application UI:** Streamlit interface where users select district, crop, month, and scenario to get advisory. [cite: 10]
- **Knowledge Base:** ICAR plans, IMD advisories, and verified ENSO literature for 4 Maharashtra districts. [cite: 10]

CORE ENGINE

- **Semantic Retrieval:** Scenario-weighted search that reprioritizes content based on El Niño intensity. [cite: 10]
- **Traceable Generation:** Every advisory structured as Recommendation → Reason → Source. [cite: 10]
- **Rigorous Evaluation:** Comparative analysis against a plain LLM baseline to prove grounding advantage. [cite: 10]

PROJECT DELIVERABLES

- **Working Prototype:** Streamlit-based web interface. [cite: 10]
- **Literature Review:** Verified analysis on ENSO-India agromet relationships. [cite: 10]
- **Code Repository:** Secure, structured, and modular source code. [cite: 10]
- **Project Report:** Comprehensive documentation of the entire research and build. [cite: 10]

ACTIVITY	WEEKS	STATUS
Literature Survey	W1–2	**Done**
Project Definition / Synopsis Approval	W1–2	**Done**
Requirement Gathering (Knowledge Requirements Matrix)	W3–4	**Done**
Dataset Collection (ICAR/IMD)	W3–5	**Done**
System Design (Architecture)	W4–5	**Done**
Core Implementation & Coding	W6–9	Planned
Mid-Term Review	W8	Planned
Testing & Debugging	W10–11	Planned
Result Analysis	W11–12	Planned
Final Report Writing	W13	Planned
Poster Preparation	W13	Planned
Final Viva	W14	Planned

REFERENCES

01 **K. K. Kumar, B. Rajagopalan, M. Hoerling,G.Bates,andM.Cane(2006)**
"Unraveling the Mystery of Indian Monsoon Failure During El Niño," *Science*,vol. 314, no. 5796, pp. 115–119. https://doi.org/10.1126/science.1131152

02 **"Regional and temporal variability of Indian summer monsoon rainfall in relation to El Niño Southern Oscillation" (2023)**
ScientificReports , vol. 13, art. 12643. https://doi.org/10.1038/s41598-023-38730-5

03 **D. S. Pai and S. Nair (2022)**
"ImpactofEl-Niño-SouthernOscillation (ENSO) on extreme temperature events over India," *MAUSAM*, vol. 73, no. 3, pp. 597–606. https://doi.org/10.54302/mausam.v73i3.5932

04-11 **Verification Pipeline**
Eightadditionalsources currently in the verification pipeline — to be finalized for Review-II.



THANK YOU

Questions & Feedback

"We are now open to questions and feedback from the committee."

SOMAIYA VIDYAVIHAR

K J Somaiya Institute of Technology
(Formerly known as K J Somaiya Institute of Engineering and Information Technology)
An Autonomous Institute Permanently Affiliated to University of Mumbai.

Somaiya TRUST

ONE-PAGE PROJECT SUMMARY

1. PROJECT DATA & EXECUTIVESUMMARY

Title: AgriAdvisor — El Niño-Aware RAG Agricultural Advisory [cite: 14]

Team & Guide: Panth Shah, Vedant Shinde, Hamood Siddique | Prof. B. Ingle [cite: 14]

RAG system synthesizing ICAR, IMD, and ENSO literature into localized agricultural advisories for 4 drought-prone Maharashtra districts. [cite: 14]

2. PROBLEM STATEMEN

The Fragmentation Challenge: [cite: 14]

Reliable officialadvisories exist but remain highly fragmented across disparate websites. No centralized system synthesizes this data into localized, reasoned guidelines for specific crops and El Niño conditions. [cite: 14]

3. KEY OBJECTIVES

Build integrated knowledge base (ICAR, IMD, ENSO) [cite: 14]

- Implement intensity-based RAG retrieval pipeline [cite: 14]
- Generate traceable (Rec → Reason → Source) advisories [cite: 14]
- Benchmark quality against standard LLM baselines [cite: 14]

4. METHODOLOGY & TECH STACK

Models: Embedding-based semantic retrieval + Local LLM [cite: 14]

Tools: Python, LangChain, Ollama, ChromaDB, Streamlit [cite: 14]

Datasets: ICAR Contingency Plans, IMD Agromet, NOAA ONI [cite: 14]

5. OUTCOMES & DELIVERABLES

Prototype: DB & District modules complete; pipeline in progress [cite: 14]

Finding: Primary gap is scenario-based retrieval, not data volume [cite: 14]

6. ADMINISTRATION & SIGN-OFF

Year: 2026–27 | **Sem:** III (Odd Sem) [cite: 14]

Guide Signature: _____ **[cite: 14]**